

Supplementary Material: Temporal Structure for Mortality Risk Stratification: Evaluating Encoding Strategies Across LLM Paradigms on UK Biobank

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1 Implementation and Reproducibility Details

Hardware and Runtime. Qwen3-8B embeddings were generated on an NVIDIA RTX 3090 (124,158 patients, ≈ 7 h per embedding run using `sentence-transformers`). Bio_ClinicalBERT (110M params) ran on an NVIDIA RTX 4090 at ≈ 30 min per pass. CoxPH models were fitted on CPU with `lifelines` v0.27; evaluation used `scikit-survival` v0.22 for TD-AUC and IBS. PCA for concatenation fusion and all CoxPH coefficients are fitted on the training split only; validation and test transforms use training-fit parameters.

Code and evaluation scripts are available at: https://github.com/Finley-Maple/Death_Clock.

Software Environment. Key package versions: Python 3.10, PyTorch 2.1, `transformers` 4.53, `lifelines` 0.27, `scikit-survival` 0.22, `scikit-learn` 1.4, `numpy` 1.26.

2 Feature Dimensions

Table 1 summarises the exact dimensionalities at each stage of the pipeline. The 39-variable count refers to raw clinical variables before any categorical encoding (1 age, 1 sex, 1 BMI, 1 smoking status, 1 alcohol status, 1 living alone, 21 blood biomarkers, 11 binary comorbidity flags). For S1, these are directly standardised and fed to CoxPH. For Settings 3 and 4 with concatenation fusion, the 39 raw features are appended to a 64-dimensional PCA-compressed embedding, forming a 103-dimensional CoxPH input. For summation fusion, the embedding is PCA-projected to 39 dimensions to match the raw feature block, then added element-wise.

Table 1. Feature and embedding dimensions by pipeline stage. d_{enc} denotes the encoder output dimension before PCA.

Pipeline stage	Setting(s)	Dimension	Notes
Raw feature block	S1, S3–4	39	Demographics, biomarkers, comorbidities
Qwen3-8B output	S3, S3-Full, S4	4096	Last-token pooling; 8192-token context
Bio_ClinicalBERT output	S3, S3-Full, S4	768	Mean pooling; 512-token hard limit
PCA embedding (concat)	S3, S4	64	Fit on training split
PCA embedding (sum)	S3, S4	39	Matched to raw feature width
CoxPH input (S1)	S1	39	Direct raw features
CoxPH input (concat)	S3, S4	103	39 raw + 64 PCA
Summation fusion output	S3, S4	39	Element-wise sum
S4 per-event repr.	S4	$128 + d_{\text{enc}}$	Age sinusoid ICD token; mean-pooled

3 Bootstrap Procedure

Bootstrap confidence intervals are computed as follows:

1. Draw 1,000 resamples of the test set ($n = 18,623$) with replacement, using a fixed seed (42) for reproducibility.
2. In each replicate, the *same* index sample is used for all methods (paired resampling), so difference CIs reflect within-replicate covariance and are not inflated by resample-to-resample variation.
3. For each replicate, compute the Harrell C-index for each method and the pairwise difference $\Delta C = C(A) - C(B)$.
4. The 95% CI is the (2.5th, 97.5th) percentile of the 1,000 difference values.
5. The one-sided p -value is the proportion of resamples where $\Delta C \leq 0$ (i.e. method A did not exceed method B).

Table 2 reports per-method bootstrap 95% CIs for the C-index. The narrow CI widths (≈ 0.017 for each method) and the extremely tight difference CIs for S4 vs. S1 (≈ 0.001 wide) both reflect the large test set. With $n = 18,623$, the bootstrap standard error of the C-index is approximately 0.004, giving a 95% CI width of ≈ 0.016 – 0.018 for individual methods; for paired differences, the within-replicate covariance further compresses the CI width.

Table 2. Per-method bootstrap 95% C-index CIs (1000 resamples, paired, seed 42). Reference configurations only.

Method	C-index	95% CI
S1 — Baseline CoxPH	0.6191	(0.610, 0.627)
S2 — Delphi zero-shot	0.6161	—
S3 — Qwen3-8B concat	0.6156	(0.607, 0.624)
S3 — BCB concat	0.6225	(0.614, 0.631)
S3-Full — Qwen3-8B	0.6195	(0.611, 0.628)
S3-Full — BCB [†]	0.6371	(0.629, 0.646)
S4 — Qwen3-8B concat	0.6233	(0.615, 0.631)
S4 — BCB concat	0.6232	(0.615, 0.631)

[†]S3-Full-BCB is an implementation-sensitive artefact (512-token truncation); excluded from headline claims (see main text).

4 S3-Full-BCB Anomaly: Metric Consistency Verification

Setting 3-Full-BCB achieves the highest raw C-index (0.6371) among all 12 cells, yet its mean TD-AUC (0.507) is near chance and its horizon-specific TD-AUC values (0.510, 0.508, 0.505) are essentially flat across 9.5–19.7 years.

To rule out a computational cause, we verified:

- **Shared evaluation loop:** all 12 methods are evaluated using the identical `scikit-survival` pipeline; time grids, censoring indicators, and inverse-probability-of-censoring weights (IPCW) are shared across methods in the same evaluation call.
- **C-index vs. TD-AUC ordering:** C-index measures overall rank agreement (discrimination), while TD-AUC at each horizon measures rank agreement within a time window. A model that ranks patients correctly on average but concentrates its predictive signal in a single feature (here: age, leaked through truncation) can achieve high C-index while producing a flat TD-AUC profile.

The hypothesised mechanism is that BCB’s 512-token limit truncates the disease-history tail, causing the input to consist predominantly of age tokens and recent high-severity events rather than the full longitudinal trajectory. The model effectively learns a strong age signal, which consistently ranks patients but carries no time-specific risk information, explaining the C-index/TD-AUC divergence. We therefore exclude S3-Full-BCB from primary conclusions and recommend longer-context clinical encoders (e.g., Qwen3-8B with 8192-token context) for Setting 3-Full.

5 Cohort and Split Details

Split sizes and event rates. The 70/15/15 stratified split yields: train 86,911, val 18,624, test 18,623 (event rate 26.2% in each split by stratification). All-cause mortality after age 60 is the binary outcome; survival time is time from age 60 to death or last follow-up date.

Feature extraction cutoff. All features are extracted from records prior to the participant’s 60th birthday. The age-60 cutoff was chosen to (a) exclude features that are themselves consequences of late-life illness (collider bias), and (b) align the prediction task with a realistic clinical decision point for preventive intervention. ICD-10 diagnoses from hospital episode statistics fields p41202, p41204, p41270 are used for the disease history.